

8-PUZZLE

```
goal = [1, 2, 3, 4, 5, 6, 7, 8, 0]
print("Enter the initial state (9 numbers):")
state = list(map(int, input().split()))
if state == goal:
    print("Goal State Reached!")
else:
    count = 0
    for i in range(9):
        if state[i] != goal[i]:
            count += 1
    print("Misplaced Tiles =", count)
print("\nInitial State:")
for i in range(0, 9, 3):
    print(state[i:i+3])
print("\nGoal State:")
for i in range(0, 9, 3):
    print(goal[i:i+3])
```

OUTPUT:

Enter the initial state (9 numbers):

1 2 3 4 5 6 0 7 8

Misplaced Tiles = 3

Initial State:

[1, 2, 3]

[4, 5, 6]

[0, 7, 8]

Goal State:

[1, 2, 3]

[4, 5, 6]

[7, 8, 0]